

Henry Tran

(617) 817-1860 | henrytran07@gmail.com | henrytran07.github.io | github.com/henrytran07

EDUCATION

University of California, Los Angeles

Bachelor of Science in Electrical Engineering

Sep. 2026 – Expected June 2028

Los Angeles, CA

Mt. San Antonio College

Associate of Arts in Mathematics

Aug. 2022 – June 2026

Walnut, CA

- **Relevant Coursework:** Object-Oriented Programming in C++, Data Structures and Algorithms, Linear Algebra, Differential Equations, Electricity and Magnetism, Modern Physics, and Quantum Physics.

EXPERIENCE

Machine Learning Researcher

Algoverse AI Research

May 2025 – Aug. 2026

Remote

- Built the core research pipeline testing whether step-level reasoning-error feedback from a stronger “teacher” model can be distilled into a smaller “student” model without any ground-truth labels.
- Designed and implemented the full pipeline end to end: offline teacher labeling, LoRA fine-tuning with a custom multi-loss objective (score MSE, feedback cross-entropy, and hidden-state alignment), and a ProcessBench-style evaluation harness.
- Architected the codebase for rapid iteration, such as modular losses, config-driven experiments, adaptive knowledge-distillation weighting, and smoke-test tooling, and now used by collaborators to run the project’s statistical bake-offs and transfer-learning experiments.

Team Lead Programmer

Mt. SAC Robotics Team

Aug. 2025 – June 2026

Walnut, CA

- Led programming of autonomous and driver-control systems; the team placed 13th of 120 at the 2026 VEX U World Championship and ranked No. 1 in California for autonomous skills.
- Built a closed-loop vision-tracking system fusing camera, distance-sensor, and odometry data through PID control, improving route consistency by 40% and earning the Think Award at the EZ-Competition (March 2026).
- Designed and built the team website used for coding tutorials and a shared knowledge archive.

PROJECTS

Fashion-MNIST Clothing Classifier | *Python, TensorFlow, Keras*

Summer 2026

- Built a convolutional neural network from scratch and benchmarked its accuracy and computational cost against a fully-connected network (also built from scratch) on a 10-class image-classification task.
- Preprocessed and normalized 70,000 grayscale images and evaluated performance using accuracy and a confusion matrix.

Chess Engine | *C++*

Summer 2025

- Designed an object-oriented chess engine from scratch using an abstract base class and polymorphic piece subclasses, so each piece type (king, queen, rook, bishop, knight, pawn) implements its own legal-move logic.
- Modeled board and piece state with `std::set` and `std::map` keyed by position via a custom hash function, enabling efficient lookups and iterator-stable updates during moves, captures, and promotions.
- Implemented full rules enforcement — legal-move validation, check/checkmate detection, pawn promotion, and draw conditions (threefold repetition, fifty-move rule, insufficient material) — in a turn-based loop with input validation and live console rendering.

TECHNICAL SKILLS

Languages: Python, C++, HTML, CSS

ML / Data: PyTorch, TensorFlow, Keras, scikit-learn, pandas, NumPy

Developer Tools: Git, GitHub, Jupyter Notebook, Visual Studio Code, Arduino IDE, VEX V5 SDK